

Efficient Dependency Resolution in IFC-aware Decentralized Programming

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Module systems are difficult!

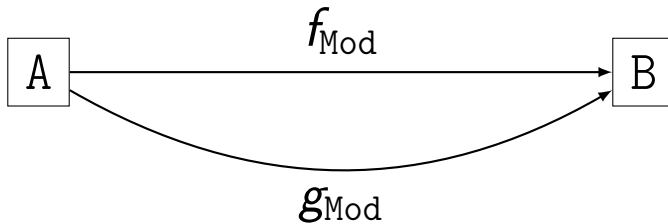
A

B

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How can one add a module cache?

How can one add a *dependency* ~~module~~ cache?

Leaky Cache² (Confidentiality)

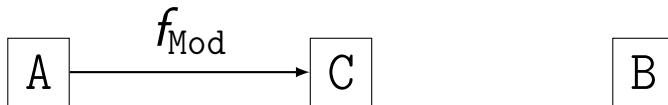
A

C

B

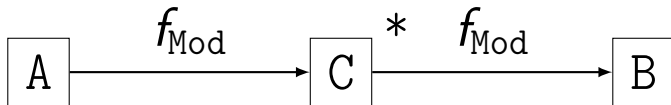
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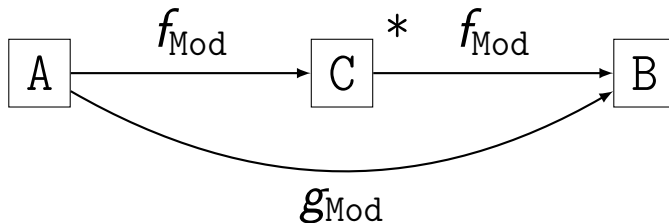
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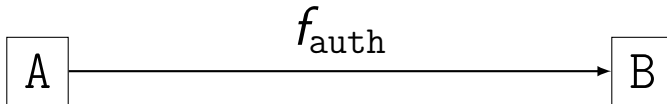
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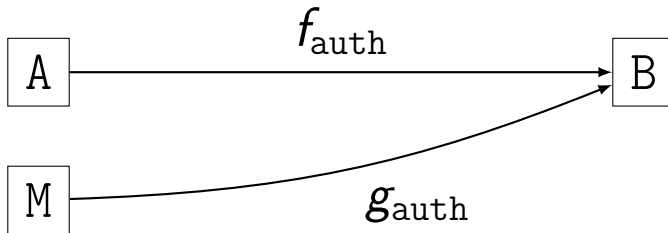
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Replay Dependency³ (Integrity)



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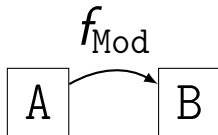
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Chain of Death⁴ (Availability)



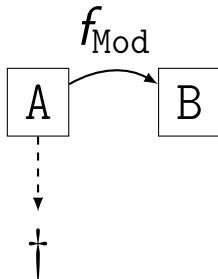
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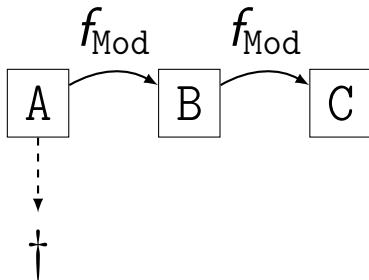
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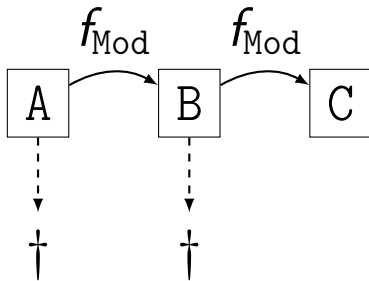
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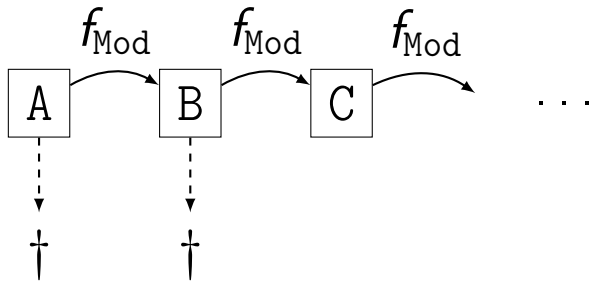
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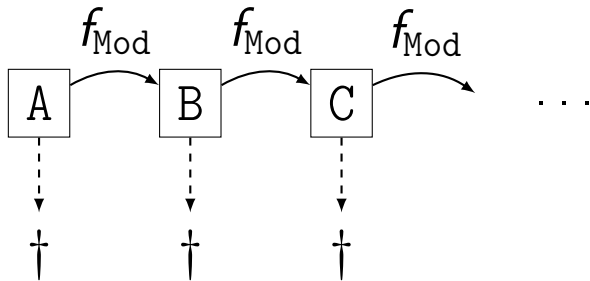
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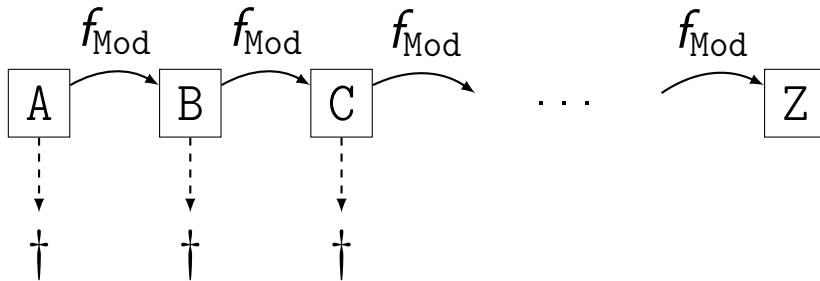
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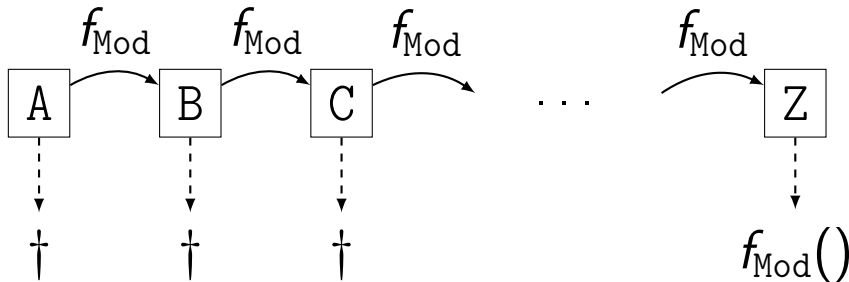
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How can one add an IFC-aware cache?

Troupe Programming Language

```
(* Alice.trp *)
```

```
1 let val msg = "Hello, Bob!"
2     fun f () = print msg
3 in send (@bob, f)
4 end
```

```
(* Bob.trp *)
```

```
5 let fun loop () =
6     receive [hn f => spawn f];
7     loop ()
8 in loop ()
9 end
```

■ Decentralized Programming

For creation of open self-organizing distributed systems.

■ Information Flow Control

Ensures via a monitor that secret information is not by accident shared with nor affected by unwanted parties.

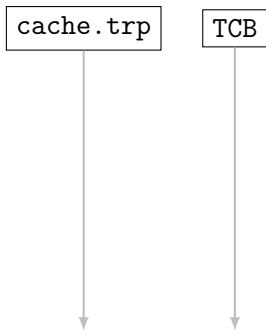
■ Dynamically Typed

Permits the dynamic behaviour needed by a truly open and updateable system.

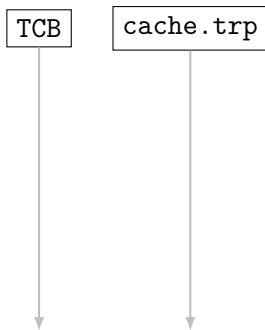
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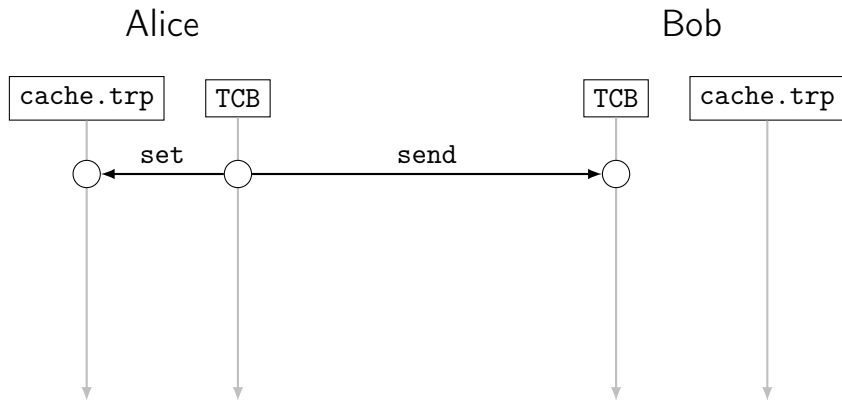
How can one add an IFC-aware cache?
(with the minimal extension to the TCB...)

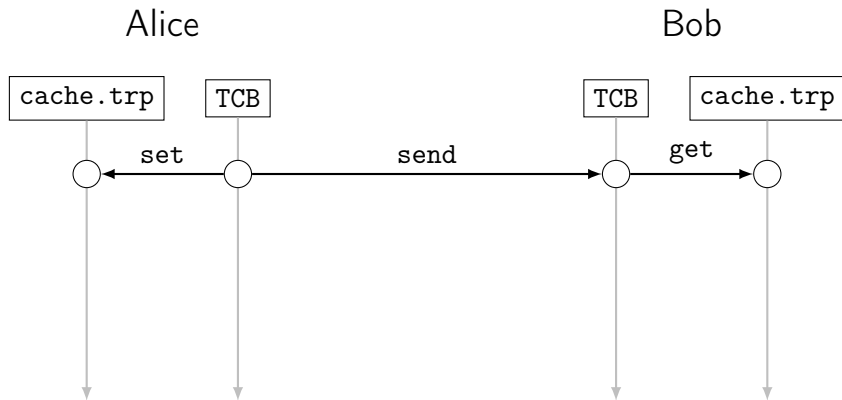
Alice

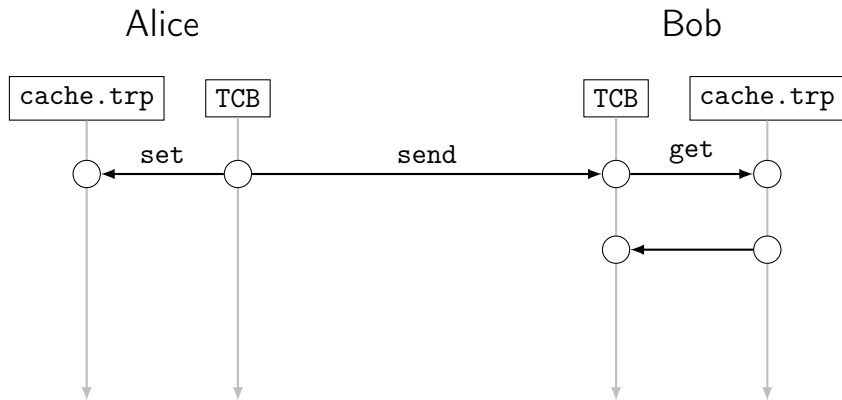


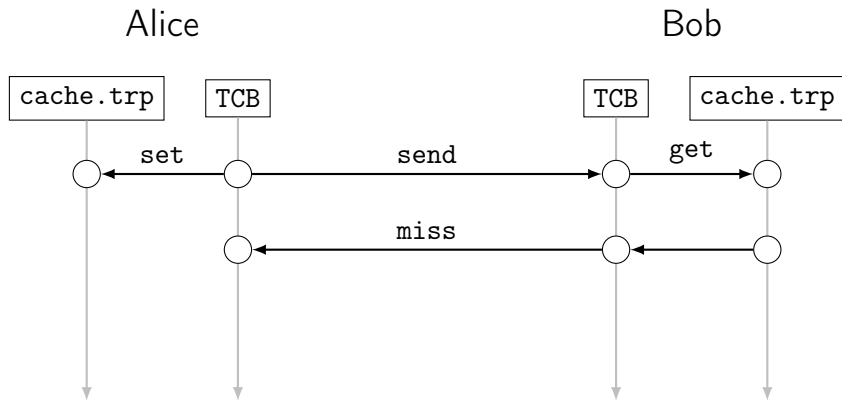
Bob

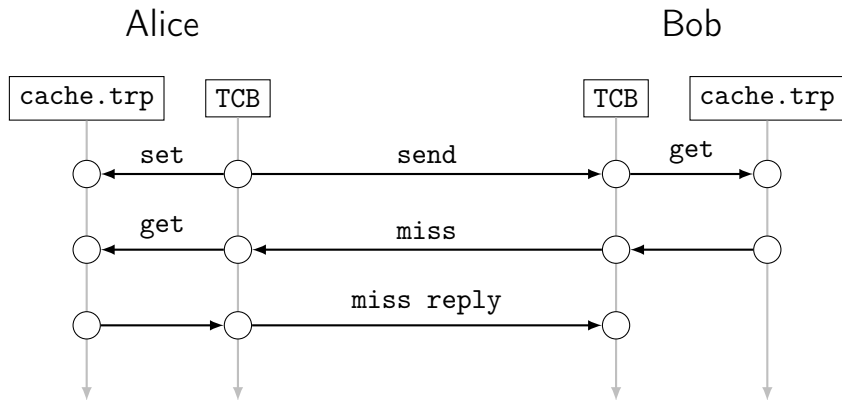


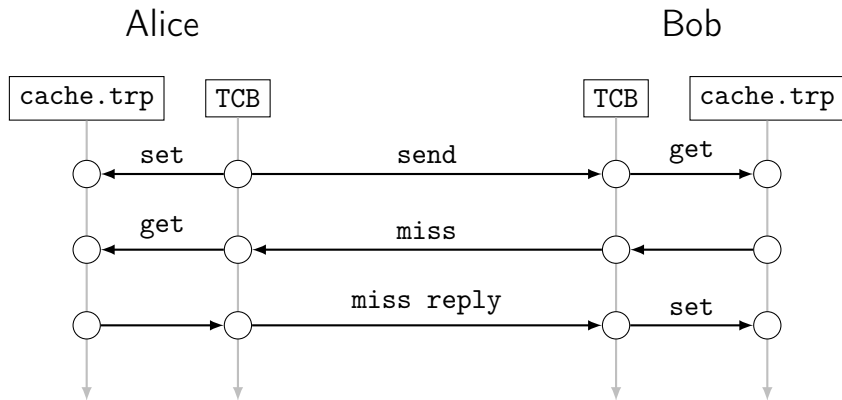












Changes to Runtime (TCB)

- **Peer-to-peer layer:**

Extend with `miss` and `miss reply` messages which are always sent by value.

- **Serialization:**

Dependency resolution requires introspection of function closures.

- During `serialize`, identify (sub)values to be referenced by hash.
- To `deserialize`, (1) extract references (2) `deserialize` given (all resolved) dependencies.

- **Hash:**

Hashing Troupe values requires introspection of function closures and source.

- **Scheduler:**

Extend to run Troupe code from the peer-to-peer layer.

Additions in Troupe: IFC-aware Map / HashMap

$$A \xrightarrow{m@{A}} B \implies \text{miss} \quad C \xrightarrow{m@{\{ \}}} B \implies \text{miss} \quad D \xrightarrow{m@{D}} B$$

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“Specification”:

An IFC-aware HashMap is a single data structure with public and secret key-value pairs which provides a IFC-aware projections of its insert and remove⁵ history:

$$i(x \mapsto 1)@{\{\}}, \quad i(x \mapsto 2)@{\{A\}}, \quad i(y \mapsto 3)@{\{B\}}, \quad r(x)@{\{B\}}, \quad i(y \mapsto 4)@{\{\}}, \quad \dots$$

⁵For this particular use case, we actually only need public (@{\}) remove operations.

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- Hash function for HashMap: SHA256 string $\rightarrow$ 32-bit int. (4 LOC)
- Define and spawn a local Troupe thread that owns the HashMap. (21 LOC)

```
(* cache.trp : line 48 *)
```

```
HashMap.insert m auth k v
```

```
(* cache.trp : line 57 *)
```

```
HashMap.findAll m auth k
```

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(* cache.trp : line 48 *)      (* cache.trp : line 57 *)  
HashMap.insert m auth k v      HashMap.findAll m auth k
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- Define and export set and get functions callable from the TCB. (8 LOC)

```
(** API Setter function *)  
fun set (key, value) = send(thread, (SET, key, value))
```

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- Use sandbox builtin to guard against timing attacks. (14 LOC)

How can one add an IFC-aware cache?

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By bootstrapping the IFC-aware runtime!

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🌐 ssoelvsten.github.io

Troupe

🔗 github.com/TroupeLang/troupe



Troupe's Threat Model

Uses several security mechanisms, inspired by Jif, Fabric, and FLAM.

- Progress-sensitive Dynamic Information-flow Control
- Sandboxing
- Capabilities

Few assumptions on adversaries! But, we do ignore:

- Network Traffic Analysis. Can be avoided with via recent methods.
(Blaabjerg and Askarov '21 & '23, Nelson, Pagnin, and Askarov '24)
- Timing-channels. Can be avoided via sandboxing and predictive mitigation.
(Zhang, Askarov, Myers '10 & '12)

Additions in Troupe: caching layer (timing-attack mitigation)

Troupe provides the `sandbox(f, t)` operation to run *f* for *t* ms and either return the result, an error, or a continuation. In practice, *t* = 1 ms seems to suffice.

```
(* cache.trp : line 48 *)  
val (m', t') =  
    sandbox' (fn () => HashMap.insert m auth k v) t m  
  
(* cache.trp : line 57 *)  
val (v, t') =  
    sandbox' (fn () => HashMap.findAll m auth k) t []
```

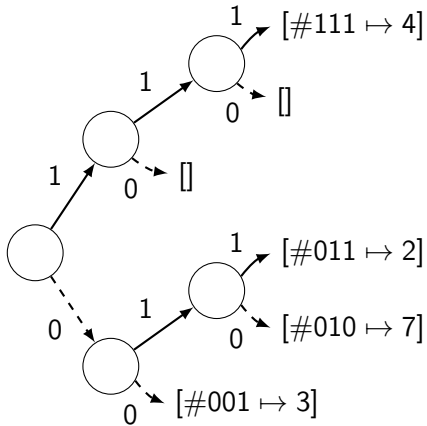

Design of the HashMap

Current prototype uses a Hash Array
Mapped Trie (HAMT) with 200 LOC.

Only 3 declassifications per insert,
remove, find (-24/+55 LOC):

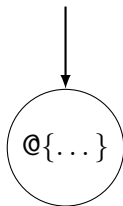
- hash key and hash bucket to not raise pc and taint the entire trie.
- The bucket list (but not its content).

But, this fully declassifies its size!



Design of the HashMap

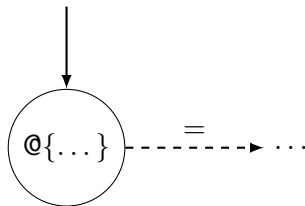
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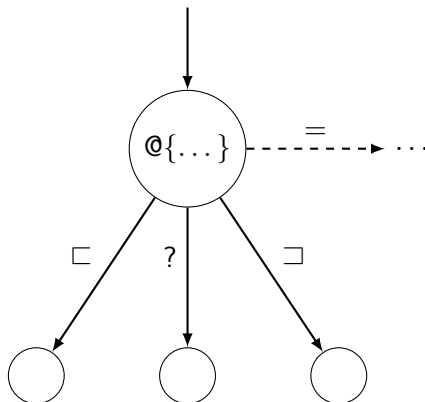
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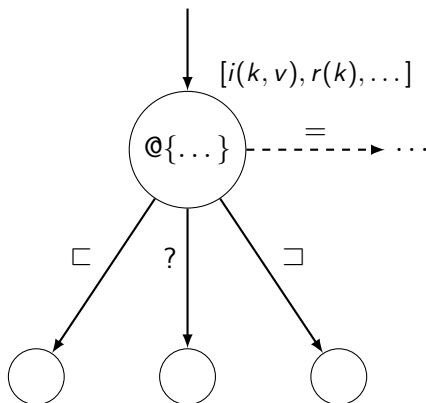
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- Use buffers to accumulate operations for “inaccessible” sub structures. This keeps `insert` and `remove` efficient.



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Can we design a meta data structure for HashMap and Map?

- Guard taint of secret sub structures with (first-class) IFC labels.
- Use the partial ordering on information flow ($\sqsubseteq$) to compare levels.
- Use buffers to accumulate operations for “inaccessible” sub structures. This keeps `insert` and `remove` efficient.
- Use time-stamps to identify latest key-value from sub structures.

But, does this make `find` too slow?

